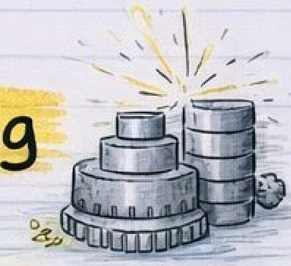
Manufacturing Machining

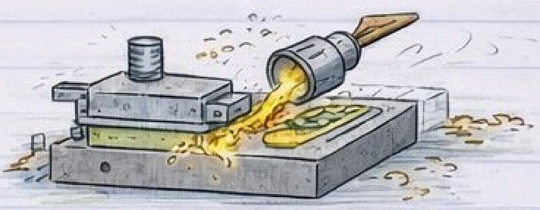
Module-1 Notes

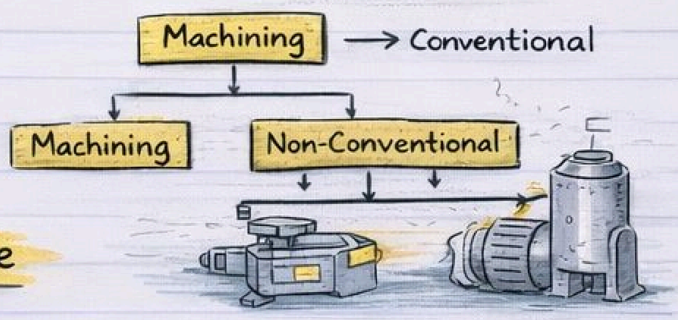
by pyqfort.com

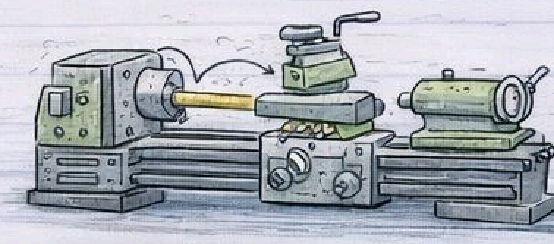


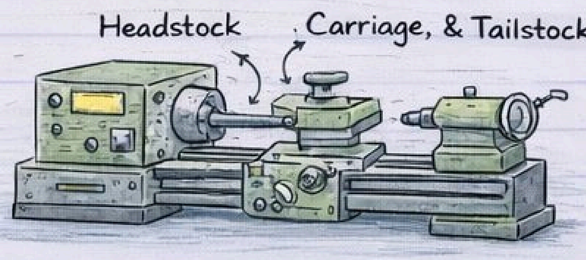
Contents Covered:

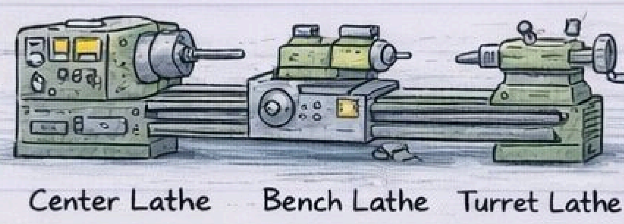
- Machining Processes: Intro
- Classification of Machining Processes
- Lathe
- Main Parts of a Centre Lathe
- Types of Lathes







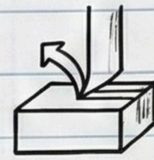




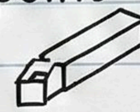
MACHINING PROCESSES: INTRODUCTION

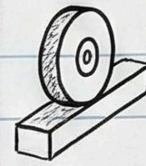
1. What is Machining?

- Gradual removal of metal from a workpiece.

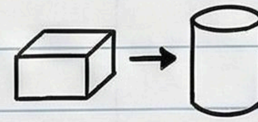


- Includes metal cutting (single/multi-point tools) & grinding (abrasive wheels).



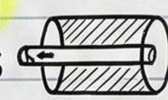


- Carried out on machine tools to achieve desired shape, size, and surface finish.



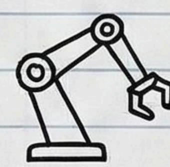
2. History & Evolution:

- First effective machine tool: horizontal-boring machine by John Wilkinson (for James Watt's steam engine).



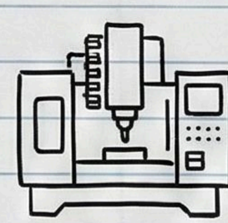
- Industrial revolution speeded up development.

- 20^{th} Century: Special purpose machines, NC tools, robots, FMS.

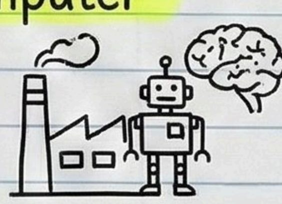


3. Modern Machining:

- Machining centre: performs several operations, automatic tool changing.

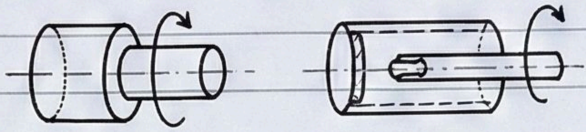


- Future: Complete automatic factory, computer-managed with robots, no workers.

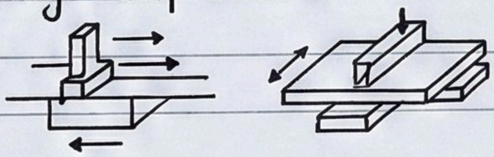


CLASSIFICATION OF MACHINING PROCESSES

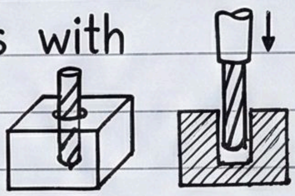
1. Turning and Boring: Turning generates external surfaces of revolution. Boring generates internal surfaces of revolution.
Tool or workpiece rotates.



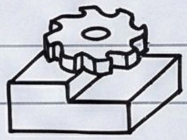
2. Shaping, Planing and Slotting: Shaping generates flat surfaces with reciprocating tool. Planing reciprocates workpiece for large flat surfaces.
Slotting is similar to shaping.



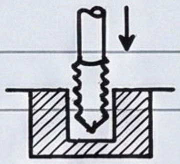
3. Drilling and Reaming: Drilling produces holes with rotary motion of tool. Reaming is a hole finishing process.



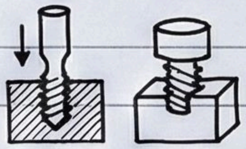
4. Milling: Generates flat and curved surfaces by rotating multi-edged cutting tools.



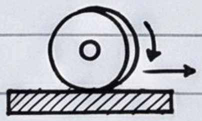
5. Broaching: Uses multi-teeth cutter for gears, key slots, internal bores.



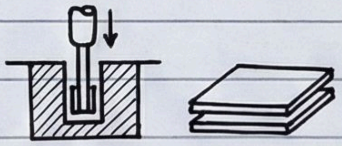
6. Thread Cutting: Tapping and die cutting produce internal and external screw threads.



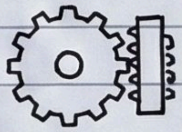
7. Grinding: Removes excess material using rotating abrasive wheel for flat and curved surfaces.



8. Fine Finishing Process:
(i) Honing (abrasive sticks),
(ii) Lapping (soft material with abrasive particles).

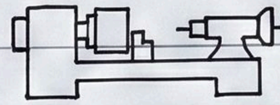


9. Gear Cutting: Milling, hobbing, shaping, rolling.
Gear finishing: Burnishing, Shaving, Grinding, Lapping.



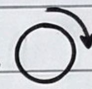
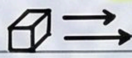
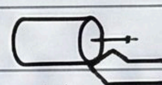
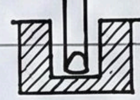
TURNING (LATHE)

1. Introduction & Purpose:

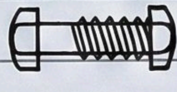
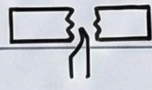


- Lathe is a basic machine tool.
- Used to machine external, internal, and face surfaces of solids of revolution. 
- Also used to cut threads. 
- Applications: Batch & mass production, repair work.

2. Principle of Turning:

- Workpiece is rotated about its axis (primary motion). 
- Cutting tool is given a feed motion normal to the cutting operation. 
- Turning generates external surfaces. 
- Boring generates internal surfaces. 

3. Key Operations:

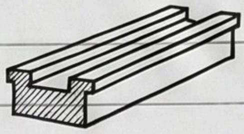
1. Cylindrical turning 
2. Facing 
3. Boring 
4. External threading 
5. Cut-off 

4. Working Mechanism:

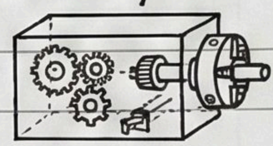
- Workpiece rotates (primary motion).
- Tool is set to desired depth of cut and moves forward (feed motion).
- Causes continuous removal of chip. 
- Resultant motion is vector sum of primary & feed motion.

MAIN PARTS OF A CENTRE LATHE

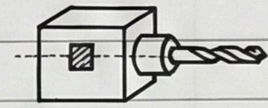
1. **Bed**:
   - Supports the main parts of the lathe.
   - Made of cast iron, consists of heavy metal slides running lengthwise with ways.



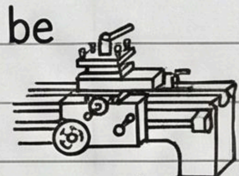
2. **Head Stock**:
   - Houses the transmission system with speed changing levers.
   - Holds devices like three jaw chuck, face plate, etc.
   - Contains hollow main spindle through which cylindrical jobs can pass.



3. **Tailstock**:
   - Movable part opposite to head stock on bed rails.
   - Supports free end of the job.
   - Holds tools for drilling, reaming, tapping.



4. **Carriage**:
   - Located between headstock and tailstock, can be locked at any position.
   - Consists of Apron (gears, clutches, lead screw), Saddle (H-shaped), Compound Rest (supports tool post), Cross Slide (transverse motion), and Tool Post (holds cutting tool).



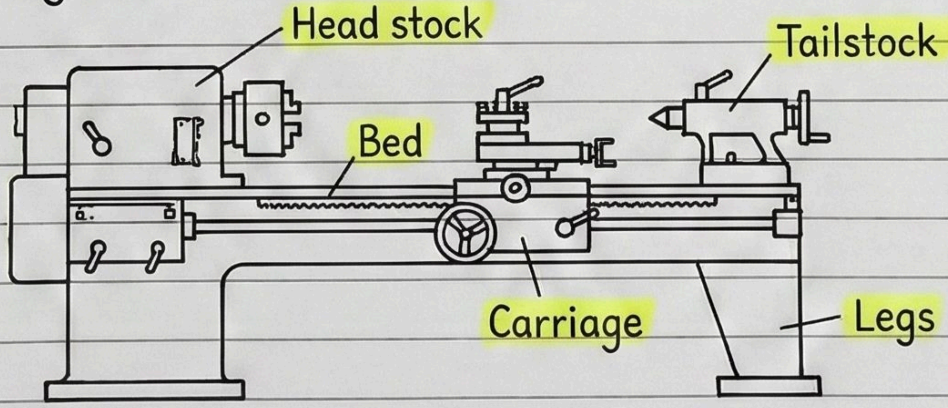
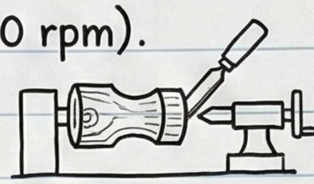


Fig. 4.2. Centre lathe

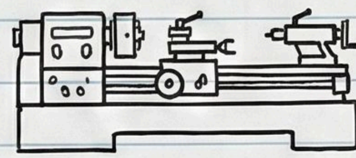
TYPES OF LATHES

Various designs and constructions to suit different machining conditions and functions. All work on same principle.

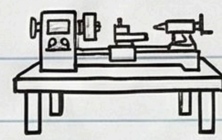
1. **Speed Lathe**: Very **high spindle speed** (1200-3600 rpm).
Simple construction (no feed box/carriage).
For wood working, centering, polishing, spinning.



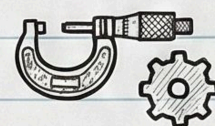
2. **Engine (Centre) Lathe**: Most **versatile** and **widely used**.
Robust headstock, feed mechanism.
Classified by power transmission: Belt-
Motor-driven
Geared-head.



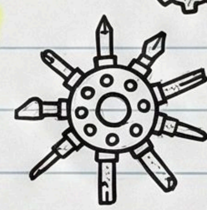
3. **Bench Lathe**: Small machine mounted on a bench.
For **small** and **precision work**.



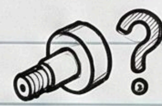
4. **Tool Room Lathe**: For **precision work** on tools, dies, gauges.
Wide speed range (up to 2500 rpm), additional accessories.



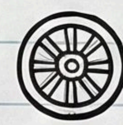
5. **Capstan and Turret Lathe**: For **mass production**.
Tailstock replaced by **hexagonal turret** holding multiple tools for sequential operations.



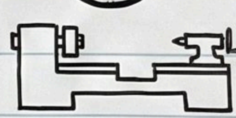
6. **Special Purpose Lathe**: For specific jobs.



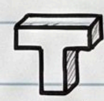
(a) **Wheel lathe**: For railroad car/locomotive wheels.



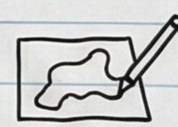
(b) **Gap bed lathe**: Removable bed section for large diameter parts.



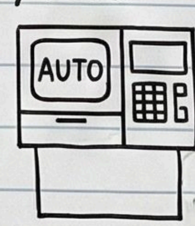
(c) **T-lathe**: For jet engine rotors, T-shape.



(d) **Duplicating lathe**: Duplicates shape of a template.



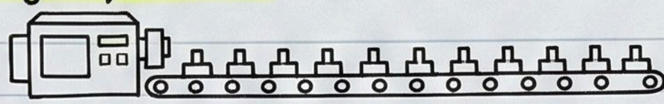
7. **Automatic Lathe**: High speed, heavy duty, **mass production**.
All operations **automatic** (start to finish, speed/feed changes, cycle repeat).



Capstan and Turret Lathe

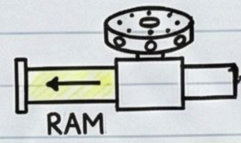
Introduction & Purpose:

- These are heavy duty, high mass production machines used for turning out large quantities of similar articles.

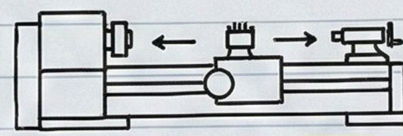


Key Differences (Capstan vs. Turret):

- **Capstan Lathe:**
  - Known as 'ram type' turret lathe.
  - Used for small and medium pieces.
  - The turret is mounted on a separate ram and its travel is restricted.

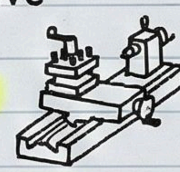


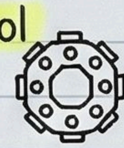
- **Turret Lathe:**
  - Known as 'saddle type' turret lathe.
  - These are large machines.
  - The turret is mounted on a slide which can operate over full length of lathe.

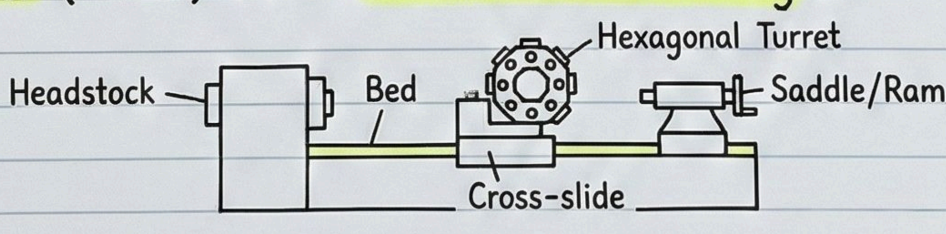


Principle Parts:

1. **Bed:** Robust box type casting with accurate guideways.
2. **Headstock:** All geared (for large lathes) or step pulley drive housing mechanisms.
3. **Cross-slide and saddle:** Equipped with a four station tool post (front) and a rear tool post.
4. **Turret saddle and auxiliary slide:** Mounts the hexagonal tool holder (turret) which indexes automatically.







Advantages:

- Tools permanently set up in proper sequence.
- Combined cuts (cross-slide & turret) save machining time.
- Multiple cuts from one tool station.
- Rigid and heavy duty for work holding.
- Many identical components machined in one setting.
- More efficient for bar work, drilling, reaming than centre lathe.